

Smart Vibration Sensor

Spring 2026 ECE 56800 Project Proposal

Vibe Coders

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ABSTRACT

Modern vibration monitoring systems are often expensive, power-intensive, and locked into proprietary cloud ecosystems, making them impractical for cost- and power-constrained deployments. We propose a low-cost, power-efficient smart vibration sensor built on the ESP32 microcontroller and MPU-6050 sensor, with total hardware costs under \$50. The system periodically wakes from a low-power state to acquire acceleration data, perform on-device signal processing including filtering and Fast Fourier Transform (FFT)-based spectral analysis, and transmit data visualizations over Wi-Fi. Targeted alerts due to anomaly detection are also reported. System applications include predictive maintenance for low-frequency industrial machinery and structural health monitoring of public infrastructure. We evaluate the system against commercial alternatives on the basis of cost, power consumption, and detection accuracy, using controlled test inputs from a signal generator and real-world validation on household appliances.

1 INTRODUCTION

The collection and profiling of vibration data is an essential process for safety critical monitoring of public infrastructure and factory machinery. As an example, significant funds are spent every year to inspect bridges [2]. Additionally, many companies frequently inspect their factory equipment by performing regular vibration testing [5]. The limitation of both of these methods is that they require the transport, setup, and teardown of inspection equipment as well as a significant investment in manual labor to conduct, verify, and report on results. Internet of Things (IoT) vibration monitoring is available from many sources to address these problems, but these solutions are high-cost [3] and often require customers to integrate into the provider's product ecosystem [?]. Because of this, data processing is handled off-chip in the cloud. An embedded IoT solution from Vibe Coders, a group of masters and PhD graduate engineering students with vibration, embedded system, and signal processing experience, offers a low cost, low barrier to entry system with on-chip processing, custom alerts, and power analysis.

2 BACKGROUND

While there are quite a few commercial options available for vibration sensing, many of them have niche applications, prohibitive cost barriers, and compartmentalized systems that require additional hardware or software to be purchased for effective monitoring. Many lower cost vibration sensors require special mounting hardware or are designed for specific machinery/equipment [1], restricting their uses to only a handful of situations. Even with lower cost devices their prices can still reach upwards of a few hundred dollars [4] while the more expensive systems easily break

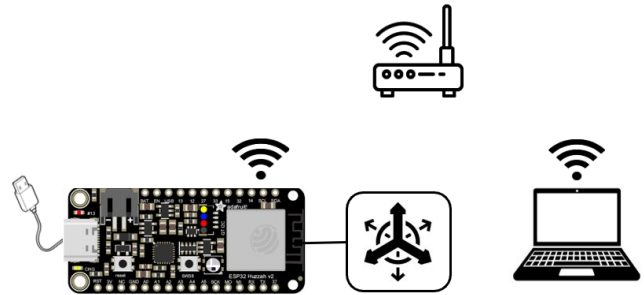


Figure 1: Concept of Operations

into the thousands [3]. In addition to this, not all sensors will provide built-in software to simplify the control and monitoring of the device, especially not the simpler systems [1]. They often require you to already have a third party monitoring software or offer one to purchase in addition to the cost of the sensor itself. Therefore, our goal is to package an "all-in-one" device which offers broad application, low cost, and built in monitoring software.

3 PROBLEM STATEMENT

Modern vibration monitoring systems often rely on expensive, high-precision piezoelectric accelerometers, designed to capture a very wide frequency range up to the kHz level for detailed fault diagnostics, and typically operate with continuous data acquisition and transmission for cloud-based analysis, making them power-intensive and complex to deploy. Consequently, they are often over-engineered for applications with strict cost and power constraints that only require basic anomaly detection or low-frequency vibration monitoring. As a result, many of such scenarios remain unmonitored.

4 PROPOSED APPROACH

To address this gap, we propose a low-cost, power-efficient, edge-based smart vibration sensor that can be easily deployed at scale. Our design, captured in Figure 1, utilizes an affordable Adafruit ESP32 Feather V2 board and an MPU-6050 accelerometer/gyroscope sensor, with a total hardware cost of less than 50 dollars. By performing on-device signal analysis and generating alerts only when necessary, the sensor eliminates the need for continuous data transmission or constant cloud connectivity. This significantly reduces power consumption and system complexity, providing a practical solution for low-cost, low-power vibration monitoring applications.

The system will periodically wake from a low-power state, and acquire acceleration data (interpreted as vibration data) over I2C. The system will then perform filtering and FFT-based spectral analysis on the collected data, and transmit the results over Wi-Fi to an online graphical user interface. MATLAB or similar will then be used for data visualization and trend/anomaly detection.

As opposed to current solutions, which utilize costly components and may require constant connection to a subscription-based cloud service, our proposed solution utilizes simple, low-cost, low-power, off the shelf components. The data collection method can easily be changed to the system of choice.

5 EVALUATION PLAN

The group intends to evaluate the functional success of our vibration detection system using two proposed methods. The first method involves testing with real-world equipment, such as a laundry machine, to assess the ability of our system to detect and report vibration patterns. The second method will use a speaker to generate configurable test patterns for our system to detect and characterize. If our system is able to accurately detect and characterize multiple vibration patterns, it will be considered a functional success. There are various signal processing elements anticipated to be necessary for our system (e.g., filters and FFT). The performance of these elements will be validated by applying known inputs from a signal generator and comparing outputs to expected results. Additionally, the system should produce configurable, meaningful alerts and data outputs.

For power performance evaluation, the group will compare power draw during active and low-power states using a digital multimeter. Measurements taken over time will allow us to quantify the efficiency of our system. Any power, cost, or feature improvements over similar commercially available systems will be considered a success.

6 TEAM QUALIFICATION

The team consists of members who, as graduate students and working engineers in industry, are experienced in more than one facet of the chosen project. We have experience in hardware aspects with FPGA/embedded, hardware design and semiconductor process engineers whose experience will be incredibly useful in building a functioning prototype. Our team-members also have the necessary experience in operating vibration testing equipment along with digital system automation. On the software side, our team has experience in digital signal processing and frequency detection algorithms, in addition to data processing / visualization which will help in building the internals of the smart vibration sensor.

7 POTENTIAL CHALLENGE

The project faces several anticipated challenges. The ESP32 and MPU-6050 impose constraints in MCU throughput, storage, and accelerometer sampling rate, which will require efficient implementation throughout and may necessitate scope reduction if needed. Coordinating seven team members across multiple sub-components will require deliberate project management, which the team will facilitate through a shared GitHub repository and weekly meetings. Finally, the absence of a calibrated vibration table complicates

performance validation; the team will instead use a speaker for controlled frequency testing and household appliances for real-world evaluation.

8 EMBEDDED SYSTEMS ASPECTS

This project uses several core concepts in embedded systems. The ESP32 communicates with the MPU-6050 sensor over I2C, a standard embedded communication protocol, and transmits processed data and alerts wirelessly over Wi-Fi. Timers and interrupts are used to schedule periodic data acquisition and trigger alert conditions, which ensures efficient use of the processor. Lastly, the project addresses resource-constrained functionality in both power consumption and computation.

9 POTENTIAL IMPACTS

The low-cost smart vibration sensor that we will develop has several advantages for monitoring low frequency machinery such as low speed fans/pumps, electric motors. A low-cost sensor lowers the barrier to entry for connecting a larger set of monitoring devices while cutting maintenance costs. Additionally, such a sensor would form a low-cost addition to smart homes in order to monitor home appliances, with flexibility in power consumption and real-time monitoring.

10 TASK BREAKDOWN AND TIMELINE

The project spans six weeks of active development, with the primary work streams consisting of: accelerometer integration (Simon), Wi-Fi communication, signal processing (Dhirodaatto, Kyle C.), low-power mode implementation (Braydon), alert system development, GUI and data visualization (Kristian, Mandy), and testing and evaluation (Kyle B.). Table 1 summarizes the task breakdown.

Table 1: Project Timeline

Week	Tasks
11 (3/23)	Team formation, project proposal, concept of operations, GitHub repository setup
12 (3/30)	Hardware setup and bring-up, I2C communication with MPU-6050, initial Wi-Fi connection demonstration, ThingSpeak feasibility investigation, preliminary filter design concepts
13 (4/6)	Low-pass filter design and implementation, FFT implementation, low-power mode investigation, alert system skeleton, GUI wireframe and planning
14 (4/13)	GUI implementation and data visualization, real-time data transmission and display, alert system implementation, initial system integration
15 (4/20)	Full system integration, testing and evaluation against defined metrics, bug fixes, final deliverable preparation, video recording
16 (4/27)	Final presentation

11 RELATION TO OTHER PROJECT

Not applicable. No direct application to other projects.

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